

# **Supplemental Materials for “Limited Degradation of Active Pharmaceutical Ingredient After Spaceflight on the International Space Station”**

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**Supplemental Methods:**

Using commercially available standards, identification of the medications was accomplished by comparing the retention time of the peak gained for each individual medication to the exact product ion obtained with tandem MS (Table S3). Linearity, limits of quantification (LOQ), carryover studies, and the precision and accuracy were then determined for each compound. Each of these are detailed below. Recovery studies were not performed because there was no biological matrix present in the standards or samples.

*Linearity*

The linearity of the method was investigated by sampling three calibrators and one quality control (QC) that were within the expected 25-100% range for each medication in triplicate. While including a zero point, quadratic fits were applied to all medications but diphenhydramine, which exhibited a linear fit. This regression analysis resulted in criteria of  $R^2 \geq 0.98$ .

*Limit of Quantification (LOQ)*

The LOQ was determined by diluting the 25% calibrator (for both promethazine and the seven combined medications set) serially until the lowest concentration that achieved linearity with the other calibrators was determined. Then six replicates of this concentration were injected and quantitated against the calibration curve. A coefficient of variation (CV) was determined for all medications. Precision in LOQ for all medications was  $\%CV \leq 10\%$ .

*Precision*

Intra-assay precision was determined by analyzing both sets of three calibrators (25%, 50%, and 100%) 4-5 times in one run. Precision at all levels was determined to be  $\%CV \leq 10\%$ . Inter-assay precision was determined by analyzing the two sets of three calibrators over five days. Precision at all levels were determined to have a  $\%CV \leq 10\%$  except for diphenhydramine at 25%

calibrator level, which had a %CV of 12%. This higher %CV was attributed to high deviance in the last batch of analysis; the criteria was still set at  $\%CV \leq 10\%$ .

### *Accuracy*

The mean accuracy was determined by comparing exact concentrations of the three calibrators to the observed signal in triplicate across five independent runs. The mean % deviation for the analytes were as follows (for 25%, 50%, and 100% levels, respectively): epinephrine (3.9, 1.5, 4.4), naloxone (7.8, 0.4, 8.0), caffeine (12.3, -2.8, -4.7), lidocaine (4.9, 1.7, 6.0), ketamine (8.5, -0.4, 8.1), diphenhydramine (1.5, 3.3, 6.8), diazepam (12.8, -3.2, 2.7), promethazine (2.7, -5.0, 1.2).

### *Carryover*

Carryover studies were conducted to determine if the needle wash solutions were sufficient to prevent carryover between injections. A 100% calibrator was injected followed by an injection of diluent (either 10% MeOH or 10% MeOH + 0.1% FA, depending on the calibrator) and compared against the LOQ. The carryover amount was determined to be near the integration threshold for all analytes (< 2%).

### *Stability*

The stability of each medication was assessed by sampling a 100% calibrator that was stored at  $5 \pm 3^\circ\text{C}$  over multiple weeks. Comparing total peak area between injections indicated modest day-to-day variability for most medications with no significant negative trend. However, epinephrine and caffeine exhibited significant declines in signal over time (epinephrine: -43.3%, caffeine: -21.7%). The degradation of epinephrine is expected due to its known conversion to adrenochrome, resulting in a pink color over time (Supplemental Figure 1). By storing the calibrator set containing epinephrine and caffeine at  $-20 \pm 3^\circ\text{C}$ , this improved stability and testing

suitability for long-term use. Additionally, promethazine exhibited high day-to-day variability but no significant negative trend. This variability was improved by addition of 0.1% FA to the original 10% MeOH diluent.

#### *Replicate Preparation*

For liquid dosage forms, if greater than 10 vials were available, one sample was removed from each vial. If less than ten vials were available, the minimum number of samples was taken from each vial to obtain 10 total chemical samples (for example, diazepam's 10 samples were obtained from three vials: three samples from vial #1, three samples from vial #2, and four samples from vial #3). For solid dosage forms, all the provided medications had at least 10 tablets or capsules. Ten tablets or capsules were randomly selected, weighed, and then either crushed into a powder (tablets) or emptied (capsules) into a clean container. From the total crushed or emptied medication, 10 samplings of approximately the same weight were then dissolved in 100 mL of diluent 2.

#### *Instrumentation Settings*

Chromatography was performed using a Waters Acquity ultra-high performance liquid chromatography (UPLC) system (Waters Corporation, Milford, MA) consisting of a temperature-controlled sample manager, binary pump, and temperature-controlled column compartment. Chromatographic separation was performed on a Waters Acquity UPLC BEH C18 (2.1 x 100 mm, 1.7  $\mu$ m) column with temperature regulated at 35°C. Mobile phases consisted of HPLC-grade water containing 0.1% FA (A) and HPLC-grade MeOH containing 0.1% FA (B). The gradient profile for all analyses consisted of 100% A for 0.5 min, followed by a gradient to 100% B over 4.5 min, and then a return to 100% A in 0.1 min, at a flow rate of 0.4 mL/min. Total chromatographic run time was 6 min. The sample manager was maintained at 20°C with a

set injection volume of 10  $\mu$ L. In order to reduce carry-over, needle washes were conducted after each injection in conjunction with diluent blank injections before and after sample batches.

Mass spectrometry was performed in tandem with UPLC using a Waters Acquity Tandem Quadrupole Detector (TDQ) (Waters Corporation, Milford, MA), operated in positive electrospray ionization mode with multiple reaction monitoring (MRM) using MassLynx software (Waters Corporation, Milford, MA). The applied capillary voltage was 2.5 kV. The source temperature was 120°C, and the desolvation temperature was 300°C. Nitrogen was used as the sheath gas and argon as the collision gas. Desolvation gas flow was set at 500 L/h, cone gas at 50 L/h, and collision gas at 0.1 mL/min. The individual cone voltages, collision energies, and other MRM parameters for each medication are described in Supplemental Table 3.



**Supplemental Figure #1:** Representative International Space Station Health Maintenance System medication storage and study packaging. (A) Representative ISS Health Maintenance System medication storage configuration. Photo credit: National Aeronautics and Space Administration. Image in the public domain. (B) Representative amber polymeric zipper-lock bags used by the Johnson Space Center pharmacy for repackaged solid oral medications included in this study. These representative images are provided to illustrate operational packaging and storage. They include (right to left) epinephrine (Ground Control – SpaceX 15), diphenhydramine (solution) (Ground Control – SpaceX 18), promethazine (tablet) (Flown – SpaceX 18), and promethazine (solution) (Flown – SpaceX 18).

| MISSION | EXPOSURE                                                                          |                                                                                    |
|---------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|         | Ground control                                                                    | Spaceflight                                                                        |
| SpX-15  |  |  |
| SpX-17  |  |  |

**Supplemental Figure #2:** Visual changes in epinephrine in both ground and control samples from SpX-15 and SpX-17. Darkened hue consistent with the formation of adrenochrome due to the oxidation of epinephrine. X-15 and X-17 refer to SpX-15 and SpX-17.

| SpX-15                     |                                 |                        |
|----------------------------|---------------------------------|------------------------|
| Medication                 | Manufacturer (MFR)              | Packaging              |
| Caffeine                   | Major                           | Amber ziplock (JSC)    |
| Diphenhydramine (capsule)  | Major                           | Blister pack (MFR)     |
| Diphenhydramine (solution) | West-Ward                       | Single-dose vial (MFR) |
| Epinephrine                | Hospira                         | Syringe (MFR)          |
| Lidocaine                  | Fresenius Kabi                  | Multi-dose vial (MFR)  |
| Naloxone                   | International Mediation Systems | Syringe (MFR)          |
| Promethazine (tablet)      | McKesson                        | Blister pack (MFR)     |
| Promethazine (solution)    | West-Ward                       | Single-dose vial (MFR) |
| SpX-16                     |                                 |                        |
| Medication                 | Manufacturer (MFR)              | Packaging              |
| Caffeine                   | Walgreens                       | Amber ziplock (JSC)    |
| Promethazine (tablet)      | Amneal                          | Blister pack (MFR)     |
| SpX-17                     |                                 |                        |
| Medication                 | Manufacturer (MFR)              | Packaging              |
| Epinephrine                | Hospira                         | Syringe (MFR)          |
| NG-11                      |                                 |                        |
| Medication                 | Manufacturer (MFR)              | Packaging              |
| Diphenhydramine (solution) | Hikma                           | Single-dose vial (MFR) |
| Lidocaine                  | Fresenius Kabi                  | Multi-dose vial (MFR)  |
| Promethazine (solution)    | Hikma                           | Single-dose vial (MFR) |
| SpX-18                     |                                 |                        |
| Medication                 | Manufacturer (MFR)              | Packaging              |
| Caffeine                   | Walgreens                       | Amber ziplock (JSC)    |
| Diazepam                   | Hospira                         | Multi-dose vial (MFR)  |
| Diphenhydramine (capsule)  | Major                           | Blister pack (MFR)     |
| Diphenhydramine (solution) | Hikma                           | Single-dose vial (MFR) |
| Ketamine                   | Mylan                           | Multi-dose vial (MFR)  |
| Lidocaine                  | Fresenius Kabi                  | Multi-dose vial (MFR)  |
| Naloxone                   | International Mediation Systems | Syringe (MFR)          |
| Promethazine (tablet)      | McKesson                        | Blister pack (MFR)     |
| Promethazine (solution)    | Hikma                           | Single-dose vial (MFR) |
| SpX-20                     |                                 |                        |
| Medication                 | Manufacturer (MFR)              | Packaging              |
| Caffeine                   | Walgreens                       | Amber ziplock (JSC)    |
| Diazepam                   | Hospira                         | Multi-dose vial (MFR)  |
| Diphenhydramine (capsule)  | Major                           | Blister pack (MFR)     |
| Diphenhydramine (solution) | Fresenius Kabi                  | Single-dose vial (MFR) |
| Ketamine                   | Mylan                           | Multi-dose vial (MFR)  |
| Lidocaine                  | Fresenius Kabi                  | Multi-dose vial (MFR)  |
| Naloxone                   | International Mediation Systems | Syringe (MFR)          |
| Promethazine (tablet)      | McKesson                        | Blister pack (MFR)     |
| Promethazine (solution)    | Hikma                           | Single-dose vial (MFR) |

**Supplemental Table #1:** Manufacturer and mechanism of packaging of each medication per re-supply mission included in the Dribble study. MFR refers to medications flown in original manufacturer packaging whereas JSC refers to medications that were re-packaged by the JSC pharmacy.

**Supplemental Table #2:** Medication calibrator and sample dilution details. Medication target concentrations (µg/mL) alongside established calibrator concentrations (µg/mL), including calibrator and sample diluents as well as dilution factors used during sample preparation. Abbreviations: MeOH, methanol; FA, formic acid.

| Analyte         | Target concentration (µg/mL) | 25% calibrator concentration (µg/mL) | 50% calibration concentration (µg/mL) | 100% calibrator concentration (µg/mL) | Diluent                    | Dilution Factor                  |
|-----------------|------------------------------|--------------------------------------|---------------------------------------|---------------------------------------|----------------------------|----------------------------------|
| Caffeine        | 40                           | 12.5                                 | 25                                    | 50                                    | 10% MeOH (#1)              | 200000                           |
| Diazepam        | 12                           | 3.75                                 | 7.5                                   | 15                                    |                            | 5000                             |
| Diphenhydramine | 0.8                          | 0.25                                 | 0.5                                   | 1                                     |                            | 50000 (liquid),<br>25000 (solid) |
| Epinephrine     | 8                            | 2.5                                  | 5                                     | 10                                    |                            | 100                              |
| Ketamine        | 4                            | 1.25                                 | 2.5                                   | 5                                     |                            | 50000                            |
| Lidocaine       | 0.4                          | 0.125                                | 0.25                                  | 0.5                                   |                            | 10                               |
| Naloxone        | 16                           | 5                                    | 10                                    | 20                                    | 10% MeOH +<br>0.1% FA (#2) | 1000                             |
| Promethazine    | 1.6                          | 0.5                                  | 1                                     | 2                                     |                            | 25000 (both forms)               |

**Supplemental Table #3:** Mass spectrometry (MS) parameters per medication. Medications were identified using commercially available standards based on parent-daughter ion pairs obtained via tandem MS. Multiple reaction monitoring (MRM) functions were developed using the retention time of each medication's peak. Cone voltages, collision voltages, and dwell times are specified for each medication.

| Analyte         | Parent ion (m/z) | Daughter ion (m/z) | Retention time (RT) (min) | Assigned MRM function (RT window) | Cone voltage (V) | Collision voltage (V) | Dwell time (ms) |
|-----------------|------------------|--------------------|---------------------------|-----------------------------------|------------------|-----------------------|-----------------|
| Caffeine        | 195.05           | 138.04             | 2.95                      | 2 (1.5 - 3.7)                     | 24               | 18                    | 106             |
| Diazepam        | 285.04           | 154.01             | 4.99                      | 3 (3.7 - 5.5)                     | 38               | 26                    | 106             |
| Diphenhydramine | 256.13           | 167.03             | 4.13                      | 3 (3.7 - 5.5)                     | 16               | 12                    | 106             |
| Epinephrine     | 183.99           | 166.02             | 0.85                      | 1 (0.6-1.5)                       | 16               | 8                     | 159             |
| Ketamine        | 238.06           | 125                | 3.16                      | 2 (1.5 - 3.7)                     | 22               | 30                    | 106             |
| Lidocaine       | 235.14           | 86.09              | 2.95                      | 2 (1.5 - 3.7)                     | 28               | 22                    | 106             |
| Naloxone        | 328.18           | 310.18             | 2.23                      | 2 (1.5 - 3.7)                     | 28               | 20                    | 159             |
| Promethazine    | 285.16           | 86.01              | 4.50                      | 3 (3.7 - 5.5)                     | 28               | 18                    | 106             |